

Model Evaluation, Explainability, and Improvement

ML Model and Foundation Model Development

03

Model choices • Training • Evaluation



Evaluation connects model behavior to business risk

A single score cannot describe every type of success or failure.

Technical quality

Use metrics that match classification, regression, generation, or retrieval.

Business impact

Measure the cost of incorrect, delayed, or unsafe outputs.

Operating behavior

Track latency, reliability, drift, and resource use.



Traditional ML metrics answer different questions

The cost of false positives and false negatives guides metric selection.

Classification

- Accuracy for balanced overall correctness
- Precision for positive prediction quality
- Recall for finding actual positives
- F1 for a precision-recall balance

Regression

- MAE for average absolute error
- RMSE gives larger errors more weight
- R-squared describes explained variation
- Business thresholds add practical meaning



Generative AI requires broader evaluation

Output quality includes meaning, safety, and usefulness.

Automated measures

BLEU, ROUGE, BERTScore, and semantic similarity compare outputs.

Human evaluation

Review correctness, clarity, relevance, usefulness, and safety.

Model-based review

LLM-as-a-judge can scale assessment but needs calibration and checks.



Explainability helps people understand model decisions

The explanation method should fit the model and the audience.

Global explanations

Describe the features and patterns that influence the model overall.

Local explanations

Describe why the model produced one specific prediction.

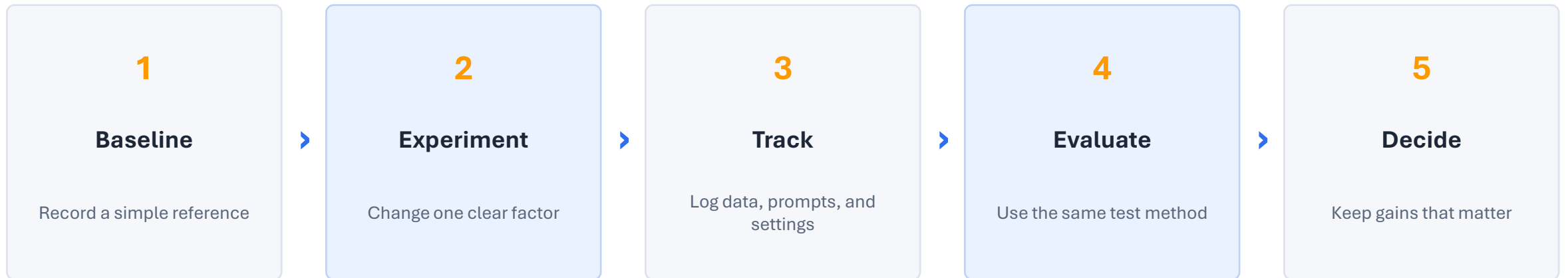
Responsible use

Check bias, misleading explanations, and limits before acting on results.



Reproducible experiments support fair comparisons

Controlled evidence makes model improvement easier to trust.



Production comparison continues after evaluation

Real traffic reveals behavior that offline tests can miss.

Shadow testing

Send copied traffic to a candidate model without affecting users.

Drift monitoring

Detect changes in inputs, predictions, or performance over time.

RAG monitoring

Measure retrieval relevance, answer grounding, latency, and failures.



Evaluation is a continuing cycle

Models, data, users, and business conditions change.

- 1 Start with a baseline and success criteria.
- 2 Choose metrics that match the task and error costs.
- 3 Evaluate subgroups, edge cases, bias, and safety.
- 4 Use human review when automated metrics miss meaning.
- 5 Explain outputs at the level users need.
- 6 Monitor drift, shadow variants, and retrieval quality after release.

